Complete parameter estimates for models testing genetic and sex-specific effects on immune-predicted weight loss (n = 3					
Model	Parameters	β	t	p	Model I
Model 1: Genetic baseline ***R^2 = 0.019, F(3,300) = 1.94, p = 0.122, AIC = 1419***		**()**			**Residua**
	Intercept	9.570 (0.411)	23.27	<0.001***	
	HI[1]	-0.154 (0.666)	-0.23	0.818	
	He[2]	-0.392 (2.035)	-0.19	0.847	
	HI × He	4.015 (3.436)	1.17	0.244	
		()			df: 300
Model 2: Main effects ***R^2 = 0.063, F(4,299) = 5.05, p < 0.001, AIC = 1407***		**()**			**Residua**
	Intercept	9.053 (0.372)	24.35	<0.001***	
	Sex (Male)	-0.090 (0.278)	-0.32	0.746	
	HI	0.185 (0.406)	0.46	0.649	
	He	1.556 (1.014)	1.53	0.126	
	Infection	**1.125 (0.285)**	**3.94**	**<0.001*** **	
		()			df: 299
Model 3: Two-way interactions ***R^2 = 0.093, F(10,293) = 2.99, p = 0.001, AIC = 1409***		**()**			**Residua**
	Intercept	9.316 (0.514)	18.14	<0.001***	
	Sex (Male)	-1.308 (0.736)	-1.78	0.077.	
	HI	-0.670 (0.828)	-0.81	0.419	
	He	1.286 (2.379)	0.54	0.589	
	Infection	**2.320 (0.795)**	**2.92**	**0.004** **	
	Sex × HI	**2.144 (0.833)**[3]	**2.57**	**0.011***	
	Sex × He	0.547 (2.076)	0.26	0.793	
	Sex × Infection	-0.369 (0.571)	-0.65	0.519	
	HI × He	1.692 (3.508)	0.48	0.630	

Significance codes: ***p < 0.001, **p < 0.01, *p < 0.05, . p < 0.1
Models compared using likelihood ratio tests.

Model	Parameters	β	t	p	Model In
	HI × Infection	-1.299 (0.841)	-1.55	0.123	
	He × Infection	-0.908 (2.095)	-0.43	0.665	
		()			df: 293

Complete parameter estimates for models testing genetic and sex-specific effects on immune-predicted weight loss (n = 3

[1] HI = Hybrid Index (0 = pure M. m. domesticus, 1 = pure M. m. musculus)

[2] He = Expected heterozygosity = 2 × HI × (1 - HI), measuring genetic admixture

[3]

Significance codes: ***p < 0.001, **p < 0.01, *p < 0.05, . p < 0.1
Models compared using likelihood ratio tests.